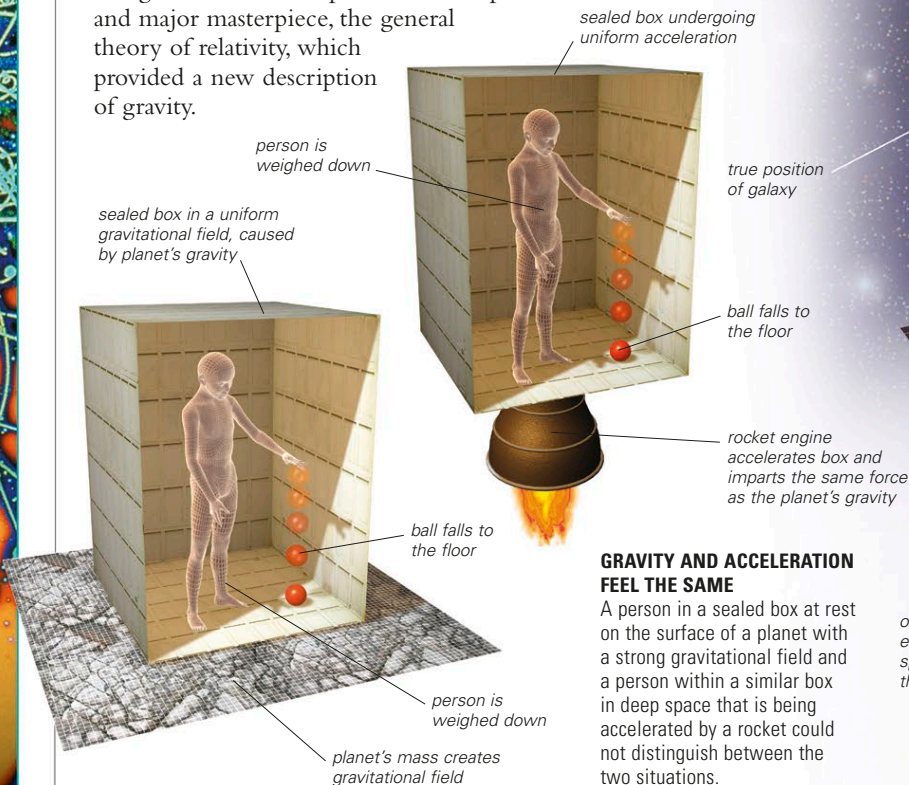


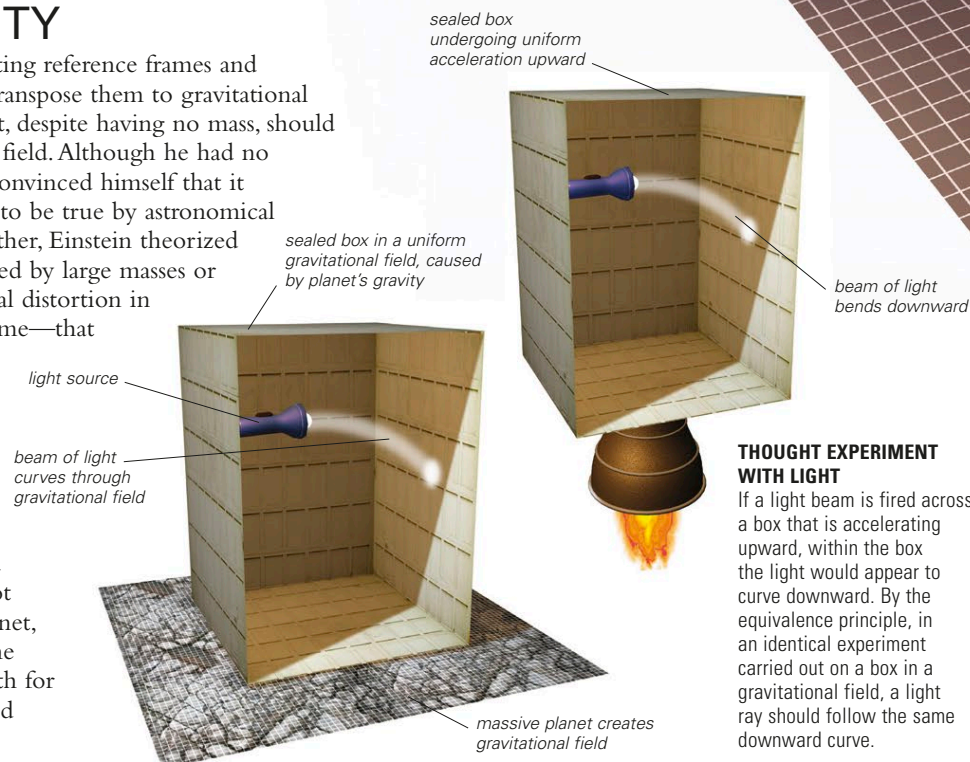
ACCELERATING MOTION

Having completed his study of relativity in the special case of reference frames in uniform motion (inertial reference frames), Einstein turned his attention to changing, or accelerated, motion. In particular, he examined the link between gravity and acceleration. This led him to formulate a proposition called the principle of equivalence, which describes gravity and acceleration as different perspectives of the same thing. Specifically, Einstein stated that it was impossible for any experiment to tell the difference between being at rest in a uniform gravitational field and being accelerated. He illustrated this idea using thought experiments involving scientists sealed into boxes under various conditions of acceleration and gravity. Starting from the principle of equivalence, by 1915, Einstein had gone on to develop his most complex and major masterpiece, the general theory of relativity, which provided a new description of gravity.



LIGHT AND GRAVITY

By visualizing experiments in accelerating reference frames and using the principle of equivalence to transpose them to gravitational situations, Einstein postulated that light, despite having no mass, should follow a curved path in a gravitational field. Although he had no direct evidence that this was true, he convinced himself that it must be. (By 1919, it had been shown to be true by astronomical observations.) Developing the idea further, Einstein theorized that gravitational effects might be caused by large masses or concentrations of energy causing a local distortion in the shape of four-dimensional space-time—that is, that gravity might be a purely geometrical consequence of the effect of mass on space-time. If so, light curves around a large mass because of the warping of space-time caused by the mass. Similarly, a planet in orbit around a star, such as the Earth around the Sun, follows a curved trajectory not because of a pull of the star on the planet, but because space-time is warped in the vicinity of the star, and the shortest path for the planet to take through this distorted region of space-time is a curved one.



orbiting planet follows elliptical path because space-time is curved in the vicinity of the Sun

two-dimensional rubber sheet represents four-dimensional space-time—dents in the sheet represent distortions of space-time

apparent position of galaxy to observers on Earth, who assume light has traveled in a straight line